

AI & Backend Achievements Report

Mini ERP Construction Management Prototype

Prepared: May 10, 2026

This report summarizes the backend and AI capabilities completed so far, focused on AI intake, proposal generation, workshop-based execution, project tracking, document generation, and API readiness.

Area	Current Achievement
AI Intake	Multilingual chat intake, hidden facts, construction prompt guidance, proposal extraction, and voice transcription.
Backend	FastAPI REST backend with SQLAlchemy persistence and APIs for ERP, AI, tracking, workshops, and documents.
Execution Model	Workshop/subcontractor-based project execution with availability, specialties, assignments, and schedule validation.
Tracking	Manual progress tracking with photos, tasks, issues, materials, dashboard warnings, and AI monitoring.
Quality	Frontend build passes, backend tests pass, and frontend-required endpoints are available.

1. Main AI Achievements

- AI Intake Chat supports Arabic, German, and English project conversations.
- Proposal Generation converts conversation context into structured project proposal data.
- Hidden Facts / Memory stores per-intake facts without confusing the normal user interface.
- Construction Domain Guidance gives the assistant renovation-specific scope awareness.
- Voice-to-Text uses AssemblyAI transcription with frontend recording/debug support.
- Proposal PDF export is available for AI-generated drafts, with Arabic handling improvements.
- AI Project Monitoring analyzes tracking context and returns risks, missing information, and recommended actions.

2. Backend/API Achievements

- FastAPI backend organized with routers and service-layer responsibilities.
- SQLAlchemy + SQLite persistence for the current prototype.
- APIs cover customers, orders, sites, workshops, invoices, work entries, payments, tracking, and AI intake.
- Project tracking APIs cover progress updates, photo uploads, tasks, issues, and materials.
- Workshop APIs cover partner CRUD, assignment to sites, schedule fields, and same-site overlap validation.
- Document endpoints support invoice PDFs/Word files, timesheets, and AI proposal PDFs.
- Postman collection and documentation exist for frontend/API testing.

3. Project Tracking And Monitoring

- Order Detail now includes a tracking center with Overview, Timeline, Photos, Tasks, Issues, Materials, and Workshops tabs.
- Dashboard displays project status, progress percentage, open issues, task completion, warnings, and upcoming actions.
- Rule-based smart warnings identify practical problems such as overdue tasks, blocked sites, high issues, missing workshop dates, unavailable workshops, and progress/status mismatch.
- AI monitoring uses the structured tracking state and rule warnings as source context, then summarizes risks and next actions.
- Arabic localization has been added for tracking labels, warnings, and AI monitoring UI.

4. Workshop-Based Execution

- The product flow was pivoted from internal employee staffing to external workshop/subcontractor execution.
- Workshop partners have name, contact, specialties, notes, active status, and availability status.
- Sites can be assigned workshops with covered trades, start date, end date, status, and notes.
- The backend blocks overlapping schedules when different workshops are assigned to the same site at the same time.
- Project tracking team view focuses on workshops and covered work instead of internal employees.

5. Quality And Readiness

- Backend tests pass: 64 tests OK.
- Frontend production build passes with Next.js.
- Frontend API usage was checked against FastAPI routes; all required endpoints are present.
- AI provider failures are handled with deterministic fallback behavior where needed.
- Environment secrets are intentionally kept out of generated documentation and source-control changes.

6. Next Planned Technical Work

- Improve UI/UX polish across Dashboard, Customers, Orders, Sites, Workshops, Drafts, Invoices, and AI Intake.
- Continue connecting project tracking with AI monitoring for better execution insights.
- Start OCR for scanned project documents such as contracts, invoices, quotations, and reports.
- Explore OCR + RAG so users can ask questions directly from uploaded project documents.
- Later: photo-based progress analysis after the manual tracking structure is stable.